



PROJECT
**ROTATING EQUIPMENT MAINTENANCE AND SERVICES –
PACKAGE 2 No. SPHR01230A**

8400211119

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TITLE

VIBRATION REPORT

JOB DESCRIPTION

MINAS GS-1

SB 1.3

08/04/2026

Approved,

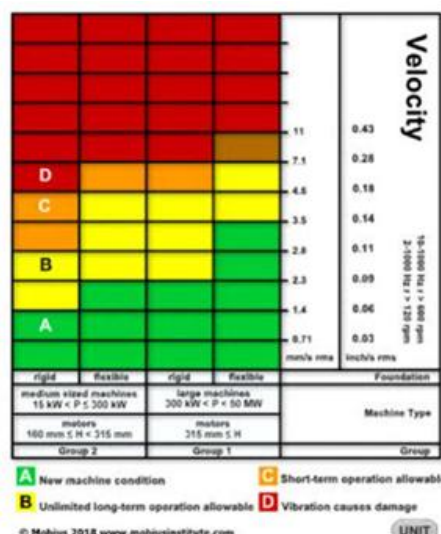
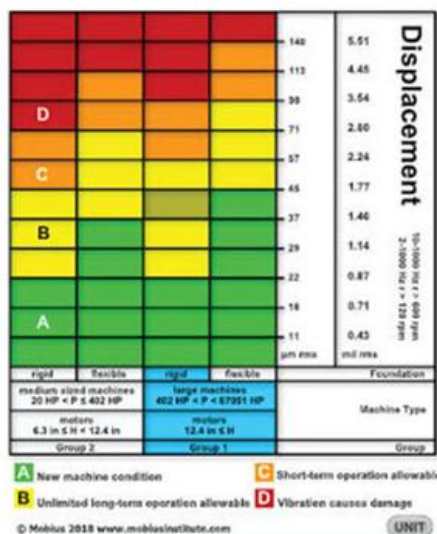
Zul Fadhlán

Spv Maintenance CS
South

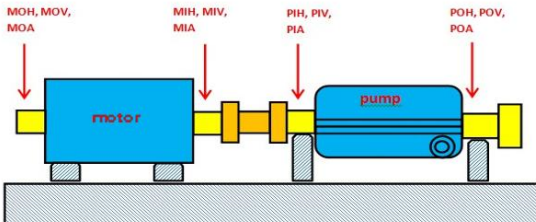
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MACHINE CLASS : CLASS 3 BASED ON ISO 10816-3		PUMP DATA	
Zone A:	Vibration of newly commissioned machine (Normal).	SERIAL NO.	: -
Zone B:	Machines considered acceptable for unrestricted long-term operation (Normal).	TAG NO.	: SB 1.3
Zone C:	Machines considered unsatisfactory for long-term continuous operation (Alert).	SERVICE	: WATER INJECTION
Zone D:	Vibration values normally considered to be sufficient severity to cause damage to the machine (Fault).	SIZE	: -
		CAPACITY	: 3000 USGPM
		TDH	: 1754 ft
		POWER	: 1750 HP
		NPSHR	: - ft
		SPEED	: 3561 RPM

ALARM LIMIT	NORMAL	ALERT	FAULT	RECENT STATUS
Overall Velocity (in/sec)	≤ 0.18	0.19-0.27	≥ 0.28	NORMAL
Overall Displacement (mils-pp)	≤ 2.9	3.0-3.9	≥ 4	NORMAL
TEMP BEARING (°C)	≤ 80	81-86	≥ 87	NORMAL



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OPERATING PARAMENTER				MACHINE CONFIGURATION
Highest Overall Vel	:	0.14	in/s RMS (MOV)	
Highest Overall Disp	:	1.8	mils-pp (MIY)	
Suction Pressure	:	7	Psi	
Discharge Pressure	:	729	Psi	
Bearing Temp M-Out	:	84	°C	
Bearing Temp M-In	:	83	°C	
Bearing Temp P-In	:	72	°C	
Bearing Temp P-Out	:	67	°C	
Flow rate	:	-	USGPM	

Finding
Overall vibration increased at the motor, with dominant 1x TS accompanied by harmonics up to 5x TS
Analysis
Spectrum indicates a mechanical issue related to the motor-pump coupling system, most likely related to coupling misalignment. However: 1. Pipe strain 2. Motor soft foot 3. Insufficient hold-down preload should be verified
Recommendation
1. Confirm motor soft foot condition 2. Confirm hold-down bolt torque against the specified value 3. Confirm pipe strain on suction and discharge piping 4. Perform shaft alignment if any mechanical deviation is found 5. Re-measure vibration after action to confirm improvement

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OVERALL LAST MEASUREMENT SUMMARY

Measurement Point	OVERALL LEVEL
MOH - Motor Outboard Horizontal	0.06 in/s RMS
MOV - Motor Outboard Vertical	0.14 in/s RMS
MOA - Motor Outboard Axial	0.12 in/s RMS
MIH - Motor Inboard Horizontal	0.05 in/s RMS
MIV - Motor Inboard Vertical	0.04 in/s RMS
MIA - Motor Inboard Axial	0.12 in/s RMS
PIH - Pump Inboard Horizontal	0.02 in/s RMS
PIV - Pump Inboard Vertical	0.01 in/s RMS
PIA - Pump Inboard Axial	0.03 in/s RMS
POH - Pump Outboard Horizontal	0.05 in/s RMS
POV - Pump Outboard Vertical	0.04 in/s RMS
POA - Pump Outboard Axial	0.04 in/s RMS
MOX - Motor Outboard X	1.3 mils-pp
MOY - Motor Outboard Y	0.9 mils-pp
MIX - Motor Inboard X	1.2 mils-pp
MIY - Motor Inboard Y	1.8 mils-pp
PIX - Pump Inboard X	0.4 mils-pp
PIY - Pump Inboard Y	0.4 mils-pp
POX - Pump Outboard X	0.7 mils-pp
POY - Pump Outboard Y	1 mils-pp



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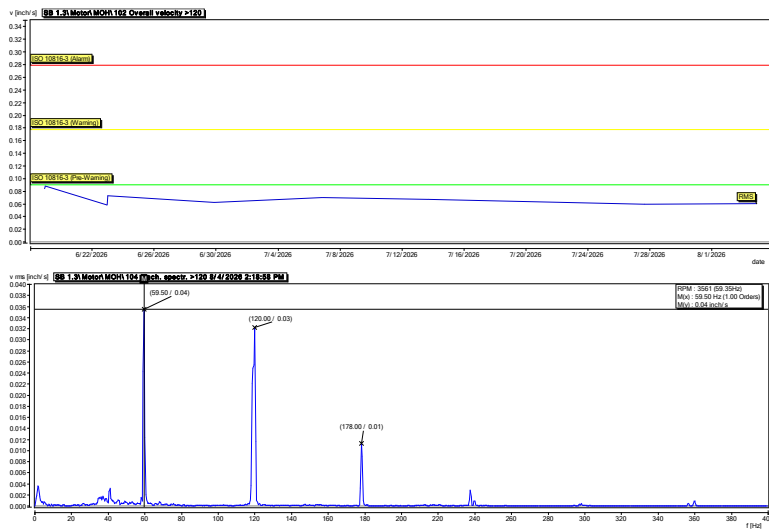
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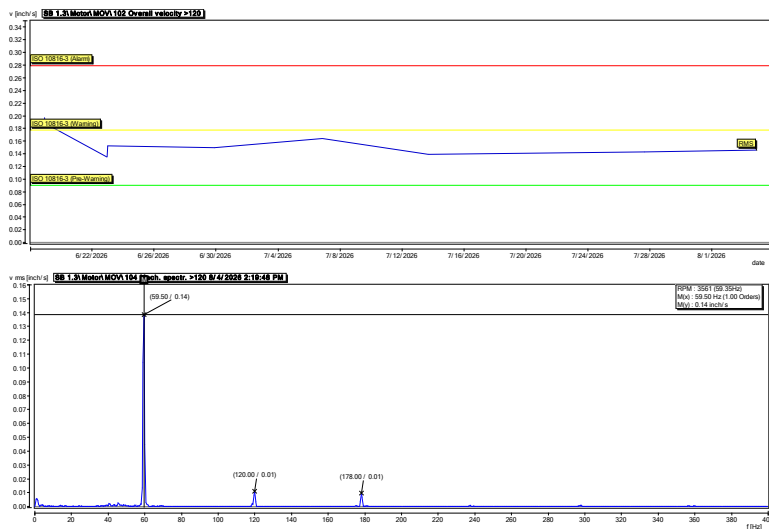
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TREND AND SPECTRUM

MOH - Motor Outboard Horizontal



MOV - Motor Outboard Vertical





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Figure 1 consists of two panels. The top panel is a plot of spectral energy density (SED) in Jy Hz⁻¹ versus frequency in GHz. It shows three components: a power-law fit (blue line), a Gaussian fit to the 100-MHz emission (yellow box), and a Gaussian fit to the 1.5-MHz emission (green box). The bottom panel is a radio spectrum plot of rms flux density in mJy/beam versus frequency in MHz. It shows three peaks labeled with their frequencies and uncertainties: (58.50 / 0.12), (126.00 / 0.02), and (178.00 / 0.03). A box in the bottom right corner provides parameters for the 100-MHz emission: P100 = 20517 Jy, S100 = 190.50 mJy, S100_Cen = 150.00 MHz, and W100 = 0.12 MHz.

Figure 1 displays two 1D NMR spectra. The top spectrum is a ^1H NMR spectrum, showing peaks at 0.50, 1.18, and 1.78 ppm. The bottom spectrum is a ^{13}C NMR spectrum, showing peaks at 0.50, 1.18, and 1.78 ppm. The x-axis for both plots is chemical shift in ppm, ranging from 0 to 400. The y-axis for the top plot is intensity in arbitrary units, and for the bottom plot is intensity in arbitrary units. The top plot is labeled "1D ^1H NMR" and the bottom plot is labeled "1D ^{13}C NMR".



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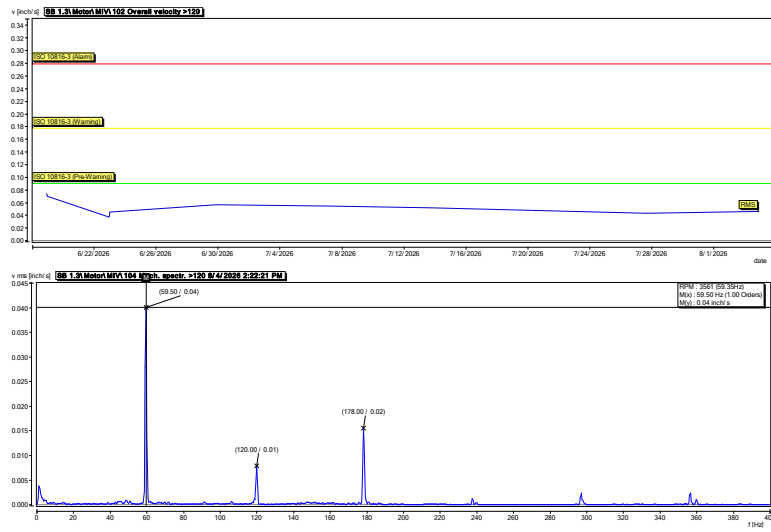


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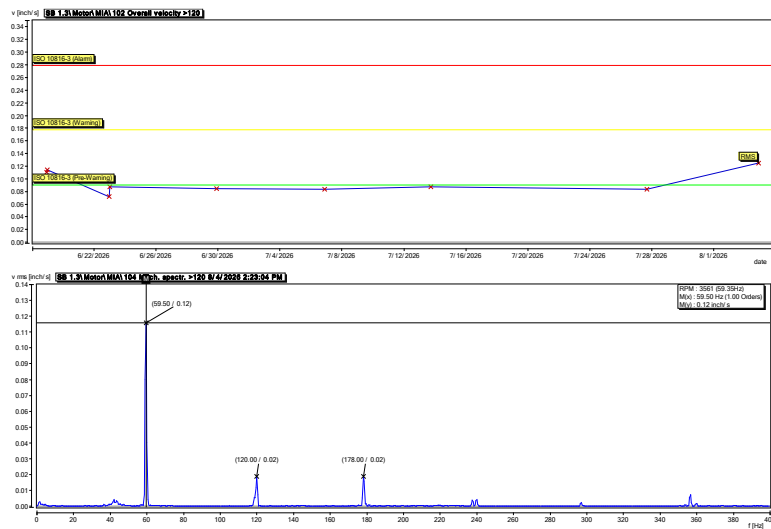
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MIV - Motor Inboard Vertical



MIA - Motor Inboard Axial





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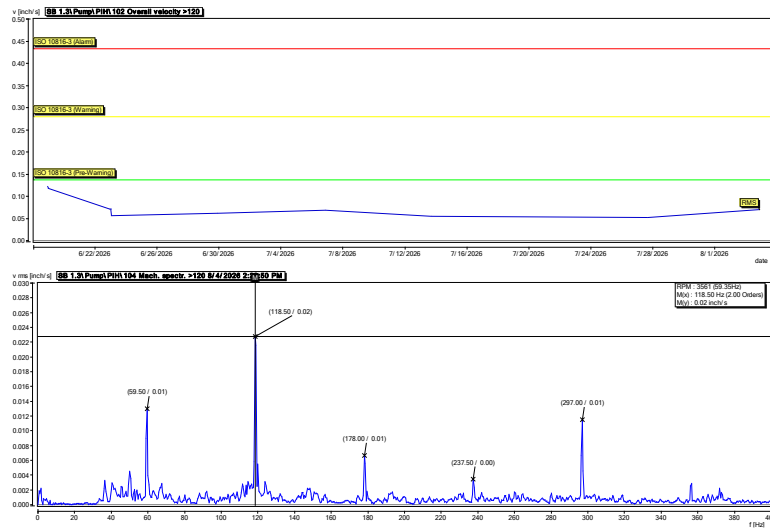


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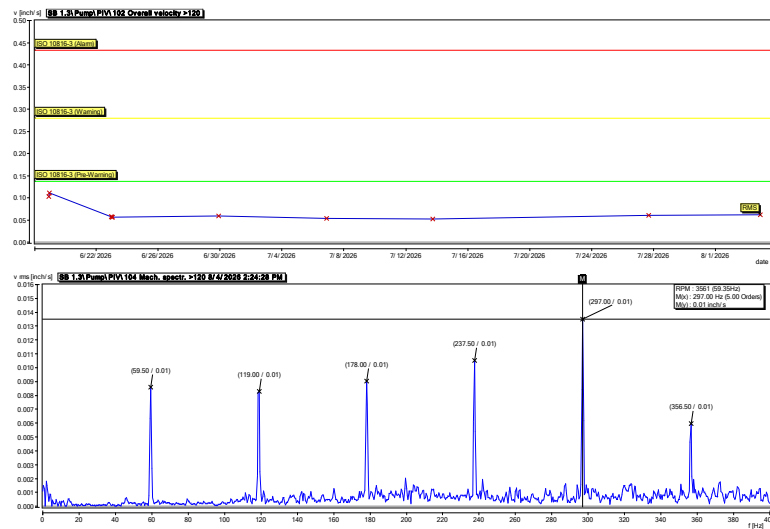
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PIH - Pump Inboard Horizontal



PIV - Pump Inboard Vertical







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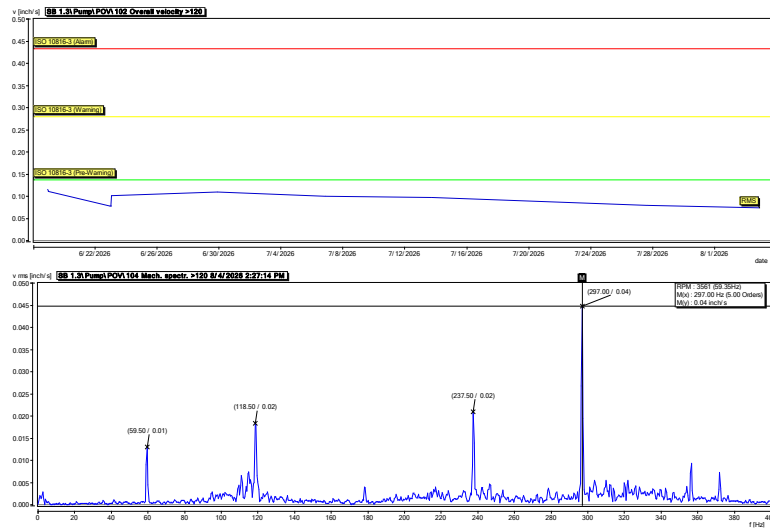


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POV - Pump Outboard Vertical



POA - Pump Outboard Axial

